How Animals Get Ready for Winter

01 How can a wood frog freeze and live?



A wood frog and a woolly bear can both freeze and live.

Look closer. Find the small frog tucked under the fallen leaves, and the fuzzy banded caterpillar resting on a leaf nearby.

A little explanation. When a wood frog starts to freeze, its liver pours out a sugar called glucose. Ice forms in its body, but outside its cells, and the glucose helps protect the cells. When the frog thaws, its heart is the first part to start again.

Try this. Draw a big circle for one frog cell. Draw ice crystals only outside the circle. Draw sugar dots near the cell. Tell how the sugar helps keep the cell safe while ice forms around it.

02 How does a chipmunk save energy all winter?



A chipmunk saves energy by letting its body cool down.

Look closer. Find the chipmunk curled up tight in a nest of leaves under the snowy ground.

A little explanation. A chipmunk is a small hibernator. Its body gets much colder on purpose. This is called torpor, and it saves energy. For most small hibernators like this one, hibernation is many long torpor stretches in a row, with short warm-ups between them. To help warm back up, many mammal hibernators burn a special fat called brown fat.

Try this. Sit very still and count your breaths for 30 seconds. Then march in place for 30 seconds and count again. Tell which time used more energy. In torpor, an animal slows its body way down to save energy.

03 How do honeybees stay warm without hibernating?



Honeybees huddle in a ball and shiver to make heat.

Look closer. Look at the round ball of bees on the comb. Find the bees on the outside and the bees packed in the middle.

A little explanation. Honeybees do not hibernate. They pack into a ball called a winter cluster. The outside bees make a blanket. The inside bees shiver their flight muscles without moving their wings, which makes heat. When there are no baby bees in the hive, the middle of the cluster is kept above about 64 degrees Fahrenheit.

Try this. Fill two jars with warm tap water and close the lids. Wrap one jar in a thick towel or some socks, like the outside bees. Leave the other jar bare. Wait 20 minutes, then touch both jars and tell which one stayed warmer.

04 Why can a mouse stay warmer under the snow?



The space under a thick snow blanket stays near freezing, not far below it.

Look closer. Find the tunnel along the ground under the snow, the mouse inside it, and the small hole that reaches up to the air.

A little explanation. The space between the ground and the snow is called the subnivean zone. About six inches of snow works like a blanket, so this space stays near 32 degrees Fahrenheit even when the air above is much colder. Voles and mice tunnel and eat there, and small air holes let carbon dioxide out.

Try this. Put two kitchen thermometers in the freezer. Bury one in a thick pile of cotton balls, like a snow blanket. Leave the other one bare. After 20 minutes, read both and tell which one is warmer.